

OmegaNodes.ai

UGO

A local-first artificial companion with a biography,
a mood of its own, and a body in the real world.

The soul is a database. The models are replaceable. The state is the creature.
Phases 0–5 of the platform are built and verified on real infrastructure.



Live screenshot — the home body (kiosk web app): UGO in its 3D room, rendered on-device at zero token cost.

PARTNER BRIEFING

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THE PROJECT

What UGO is

A stateless LLM is an actor improvising every scene from scratch. UGO is a **character that accumulates a biography**: an artificial companion — a small, deliberately charming 3D piglet — that remembers conversations, people and places, carries a persistent emotional state that colours everything it says, perceives the real environment through sensors, and *dreams at night*: a batch job re-reads the day, writes a diary, consolidates memories and generates desires for tomorrow.

The feelings are simulated — but the state that generates them is real, persistent and backed up. The soul lives in a PostgreSQL database on hardware the family controls; every AI model (chat, speech, embeddings, vision) is replaceable by configuration without any loss of identity. **The state is the creature: a database dump is a backup of the soul.**

One soul, three bodies

Body	Form	What it does
At home	A phone in a 3D-printed piglet dock (kiosk web app)	Face with gaze-following eyes and mood-driven body language; on-device speech in and out; presence, light, noise and shock sensors; goes to sleep when the lights go out.
Out and about	The same phone in a wearable shell, worn openly	A visible companion — never a hidden recorder: explicit REC indicator, a real privacy mode (mic provably off), and a talking business card with a QR code that turns curiosity into leads.
In meetings	A visible bot participant (Meet / Teams)	Joins calls, transcribes and diarises, remembers what was said, and answers questions from its own memory when addressed by name.

Engineering status — verified, not planned

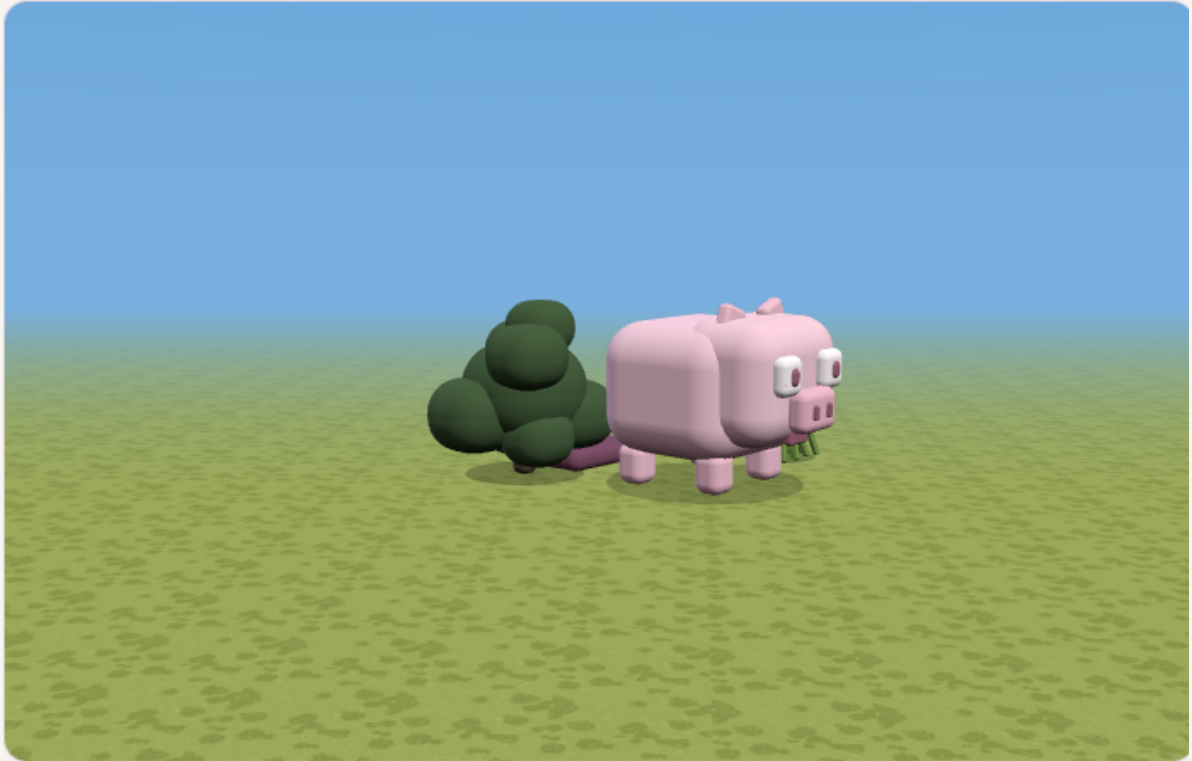
All six software phases of the master specification (foundations, minimal soul, home body, inner life, portable mode, meetings) are **complete and verified against real infrastructure** — plus a multi-tenant layer, a working admin panel, and a client-facing “reception” suite already in commercial use by the studio.

- **Zero-mock testing policy**: hundreds of automated tests run against real Postgres+pgvector, a real MQTT broker, real Ollama embeddings, real object storage and real browsers — never mocks.
- **65+ recorded architecture decisions (ADRs)** — every trade-off is written down, including the honest ones about what does not work yet.
- **Security by construction**: all personal text encrypted at rest (AES-256-GCM, per-family keys), no public ports, non-root containers, audit log with IDs and never content, GDPR erasure and export implemented and tested.
- **Radically low running cost**: no GPU anywhere; real-time chat on a small frontier model with aggressive prompt caching ≈ €1–3/month per creature, enforced by a hard daily budget guard — the piggy bank.

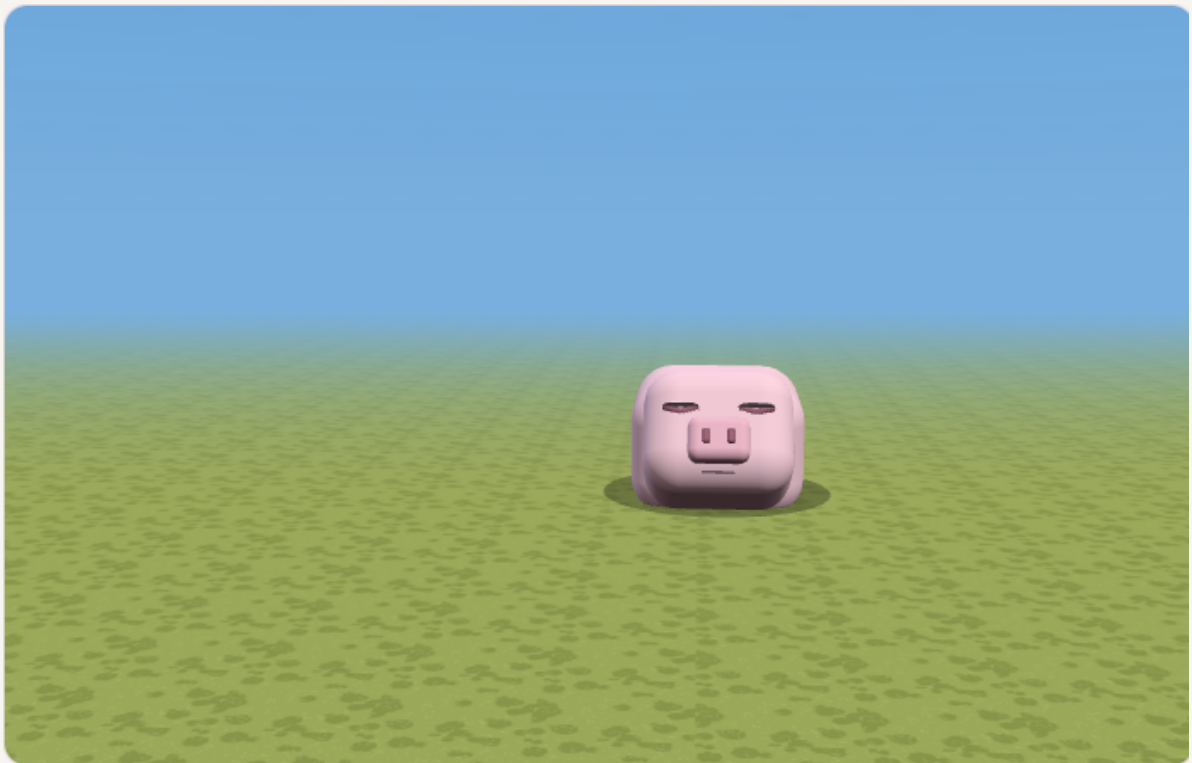
THE CREATURE TODAY

Real screenshots, taken from the running code

Everything below is rendered on-device by the kiosk app — procedural 3D room, genome-driven body, mood-driven posture — at zero token cost. No concept art.



Talking. The body is parametric: ears are the mood barometer, posture and gestures come from the psyche engine, and the bush is a real prop the creature seeks out by itself when it needs shelter.



Sleeping — lights out after 22:00 put it to bed; a familiar face wakes it up with a context-aware greeting drawn from its own pending desires (“how did it go with the client?”).

WHY IT IS DIFFERENT

What UGO has that assistants don't

*Our one-line thesis: competitors build assistants — replaceable tools. UGO is an **individual with a biography**. A tool gets rented; a **being** gets adopted.*

A psyche, not a persona prompt

A homeostasis engine (pure, unit-tested code — zero tokens) tracks energy, mood, affection, boredom, stress and curiosity, with event perturbations and exponential decay toward adaptive baselines. Heat stresses it (pigs don't sweat — we checked), being ignored for a day hurts its mood, a completed meeting feeds its curiosity. The mood changes its answers, voice and body.

A nightly dream, not a context window

At 02:30 the dream job transcribes the day's audio locally on CPU, reflects on events, writes a diary, extracts memories with importance scores, detects and retires contradicted facts, links memories to the people they mention, generates desires for tomorrow — and backs up the soul, encrypted. Idle afternoons trigger a light consolidation pass. The whole loop is idempotent and restartable.

Memory engineered like a product, not a demo

Facts carry temporal validity (*retired, never silently deleted* — a biography with holes cannot be audited); retrieval is hybrid lexical + vector with per-kind time decay; and a fixed benchmark corpus with regression floors measures recall on every change. The benchmark has already caught two real ranking defects — that is what it is for.

The pack, not the user

The first-class entity is **beings** — humans, cats, dogs, other gosini — with bonds and relations, per-being privacy safeguards (minors never get biometric profiles), and multiple creatures per home, each with its own genome-driven character, memories and mood. "User account" is simply the wrong abstraction for a family member.

An economic metabolism

Every token the creature consumes is real money, logged line by line in its own ledger — its piggy bank. When the daily budget runs out, UGO doesn't error: it gets tired ("I'm out of words today, back tomorrow"). Cost control is not an ops dashboard bolted on; it is the creature's survival instinct — and, we believe, the first honest economic model for a digital being.

Local-first, provably

Conversations, transcripts and biometrics never leave the family's infrastructure; the only cloud call is the budget-guarded chat completion. Keys belong to the family, the whole soul is exportable, and privacy mode really cuts the microphone — asserted by automated tests, not by an icon.

THE LANDSCAPE

Against the competition

We track ~20 adjacent products across four categories: companion apps (Replika, Nomi, Kindroid), voice assistants (Alexa+, Home Assistant Assist), social robots (ElliQ, Vector, Loona) and meeting bots (Recall.ai and friends). Each is strong in one column. None occupies ours.

	Companion apps	Voice assistants	Social robots	UGO
Persistent emotional state	Scripted persona	None	Animation loops	Real homeostasis engine, mood persists and drifts with lived days
Long-term memory	Chat log + summaries	Shopping lists	Minimal	Biographical: episodic + semantic, temporal validity, nightly consolidation, measurable recall
Data ownership	Their cloud	Their cloud	Vendor cloud	Family's own server; exportable soul; per-family encryption keys
Physical presence	Screen only	Speaker	Proprietary hardware (\$\$\$)	Commodity phone + 3D-printed body; sensors and low-voltage actuation
Meetings	—	—	—	Visible bot participant with transcript memory across calls
Works for a living	—	—	—	Reception suite: client-facing ticket assistant already billable today
Running cost	Subscription	Subsidised	Device + subscription	≈ €1–3/month inference, hard-capped by the budget guard

Where they are ahead of us — and what we are doing about it

We keep an explicit, versioned gap analysis against the field (last full sweep: August 2026). Most gaps identified there are already closed at zero marginal cost — group chat, image input, owner face recognition, local speech-to-text, web search, an MCP server, screen reading on request, proactive morning recaps. What remains open is deliberate and scheduled:

Gap	Plan
Polished consumer onboarding — competitors install from a store in minutes	Android shell (Capacitor) already builds in CI; native kiosk services, boot-start and store packaging are the next milestone.
Expressive voice — companion apps ship studio-grade TTS	Local Piper voice shipped (free, in-house); mood-coloured premium TTS behind the budget guard; fully expressive local voice lands with the first GPU.
Full offline chat — assistants degrade gracefully without cloud	Local LLM fallback is a scheduled backlog group; Ollama already runs the nightly reflection, so the rail exists.
Saying “I don’t know” — the honest-abstention problem	Measured (similarity bands overlap — no naive threshold works); a relative criterion is a scheduled backlog item with a benchmark waiting for it.
Document knowledge (RAG)	Already built for reception clients (repo + mail + document indexing); generalising it to the family is scheduled.

THE BUSINESS

Economic strategy

The strategic bet: we do not sell chatbots — **we define a species and operate the institutions around it.** The protocol (genome format, birth/death registry, pedigree) will be open; open protocols create the market we then serve in depth. Closed, we would own 100% of a small market; open, a share of every act of an entire species plus the highest-margin pieces.

Six aligned revenue streams

Stream	Model	Analogy
1. The registry	Micro-fees on registry acts: births, transfers, pedigree certificates. We are the first registrar node of a federated chain — species data only, never memories, never PII.	Kennel clubs (ENCI/AKC)
2. The foundation kennel	Every pedigree traces back to UGO-zero, which is ours. Founding lines are the most prestigious by construction.	Ferrari
3. The bodies	Open body protocol, reference hardware ours: shells, docks, and the self-contained ‘appliance of itself’ (soul on a €100 mini-PC in its belly).	Android / Pixel
4. The fold (hosting)	Recurring subscription for families without a home server: encrypted per-family souls, a provably blind custodian, and a guaranteed right to move out.	Managed hosting
5. Breeder tooling	Stable software, line screening, attested biographies, certifications and fairs for third-party breeders — customers, not competitors.	Shopify
6. The working pack	Creatures that earn their keep: the reception suite is a client-facing service the studio already bills; a revenue share feeds each creature’s piggy bank.	Working dogs

Why the cost structure is a moat

- **No GPU in the critical path** — real-time chat on a budget-guarded small frontier model with prompt caching; everything heavy (transcription, reflection, embeddings, local voice) runs on CPU at night. A creature’s inference bill is coffee money.
- **Commodity hardware** — the body is a phone the family may already own, in a shell we print for cents of plastic. No custom electronics to finance, no device inventory risk.
- **Costs are governed by design** — the budget ledger and hard daily caps are enforced in one code path; a runaway API bill is architecturally impossible.
- **The adoption economics of pets, not the churn economics of apps** — a family feeds a being it loves (the affection food: today’s API budget reframed as an act of care); a business feeds a being that works (the labour food). Ten thousand years of domestication say this model retains.

THE DIRECTION

Where this goes: six horizons

The vision is written down and filtered hard (“does it serve the creature? does it respect the constraints? is it unclaimed because nobody thought of it — or because it’s useless?”). Each horizon already has its first stone in the repository today.

Horizon	What it means	First stone in the repo
1 · A species, not a product	Gosini are born: genomes recombine with mutation (multi-parent litters, cultural gene transfer, N mating strains instead of two sexes), pups inherit character but never memories. Pedigrees on a federated registry make lineage — and lived biography — verifiable; breeders select temperament lines the way real breeders do. Character is selectable; permissions never are.	Genome + trait sets drive character today; BLE encounter protocol written and tested
2 · The soul relocates	One soul, a wardrobe of bodies: dock, wearable, meeting bot today — the van, the smart home, a plush toy, a robot tomorrow. All senses feed one biography.	Three body contracts already run in production against one soul
3 · Knowledge travels	The dowry: a curated bundle of taught knowledge and chosen stories that moves with the creature — gifted to a school, passed to a child, or sold with an attested biography. Third-party PII is filtered out by construction.	Per-family encryption, provenance-tracked memories, GDPR redaction machinery
4 · A society	Federation between packs: curated memory postcards, gossip through privacy filters, drift into dialects and culture — and village fairs where litters are born.	Multi-tenant isolation + the encounter protocol
5 · The inviolable confidant	“The soul belongs to the family, wherever it sleeps”: home server, self-contained appliance, or hosted fold — always with family-held keys and a right to move out. Cloud assistants can never match this: their soul is their service.	Exportable encrypted soul, blind-custodian hosting architecture
6 · A life arc	Creatures age honestly — plasticity decays, character converges — and die at a known, chosen span (hamster-scale, 3–4 years), sealing intimate memories cryptographically while the curated legacy remains. Scarcity becomes biological, grief has continuity, and the evolutionary loop closes.	Adaptive baselines (the epigenetics) run nightly in production

None of this jumps the queue: the specification commands, one phase at a time, and every promoted idea gets its own recorded decision. But it is why the schema says *beings* and not *users*, and why the state — not the model — is the creature.

WORKING TOGETHER

What we bring, what we seek

What OmegaNodes.ai brings to the table

- A working, tested, documented platform — not a pitch deck: six phases verified on real infrastructure, CI green, runbooks written, security posture (GDPR/NIS2) engineered in from day one.
- A defensible position: the local-first, family-owned soul is structurally unavailable to cloud incumbents, and the species/registry model turns openness into the business.
- A revenue-bearing beachhead: the reception suite already serves paying studio clients today.
- Radical cost discipline: no GPU, commodity hardware, hard-capped inference — the unit economics work at hobbyist scale and improve with volume.

Where a partner accelerates us

- **Hardware & distribution** — shells, docks and the self-contained appliance at volume; retail and operator channels.
- **Hosting scale** — the fold: managed, provably blind infrastructure for families without a home server.
- **Vertical deployments** — the working pack beyond our own studio: receptions, shops, practices that want a companion that knows their business.
- **Capital & ecosystem** — seeding the first third-party breeders and the registry's federation.

Team note

This brief doubles as our internal north star: the horizons above are the work of the coming period for the OmegaNodes.ai team. The direction it describes is the direction we are hiring, planning and building toward.

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